

Smart EV Charging Prediction & Recommendation

Three machine-learning models forecast a 20-station charging network one hour ahead.
A linear program then decides how to spread that load across the grid.

20

charging stations across the city

168

chargers · 11.1 MW installed

175,200

station-hours of history

297,856

charging sessions simulated

● **Driver** — where do I charge, and how long will it really take? ● **City operator** — what will the network and the grid do tonight?

No public dataset exists — so the city was simulated

A **discrete-event simulator** — not random numbers. Every car is an event with an arrival time, a queue and a charging session.

- **Eight venue types**, each with its own hourly demand shape — government, business, residential, retail, transit, leisure, hotel, education
- **Friday–Saturday weekend** and 17 Egyptian public holidays
- **Charging physics** — pack size, state of charge, power taper above 80 %, thermal penalty in cold and heat
- **M/M/c queue with abandonment** — 6.9 % of drivers give up and leave
- **Weather, venue events, +45 % adoption growth** across the year

11,082

MWh delivered

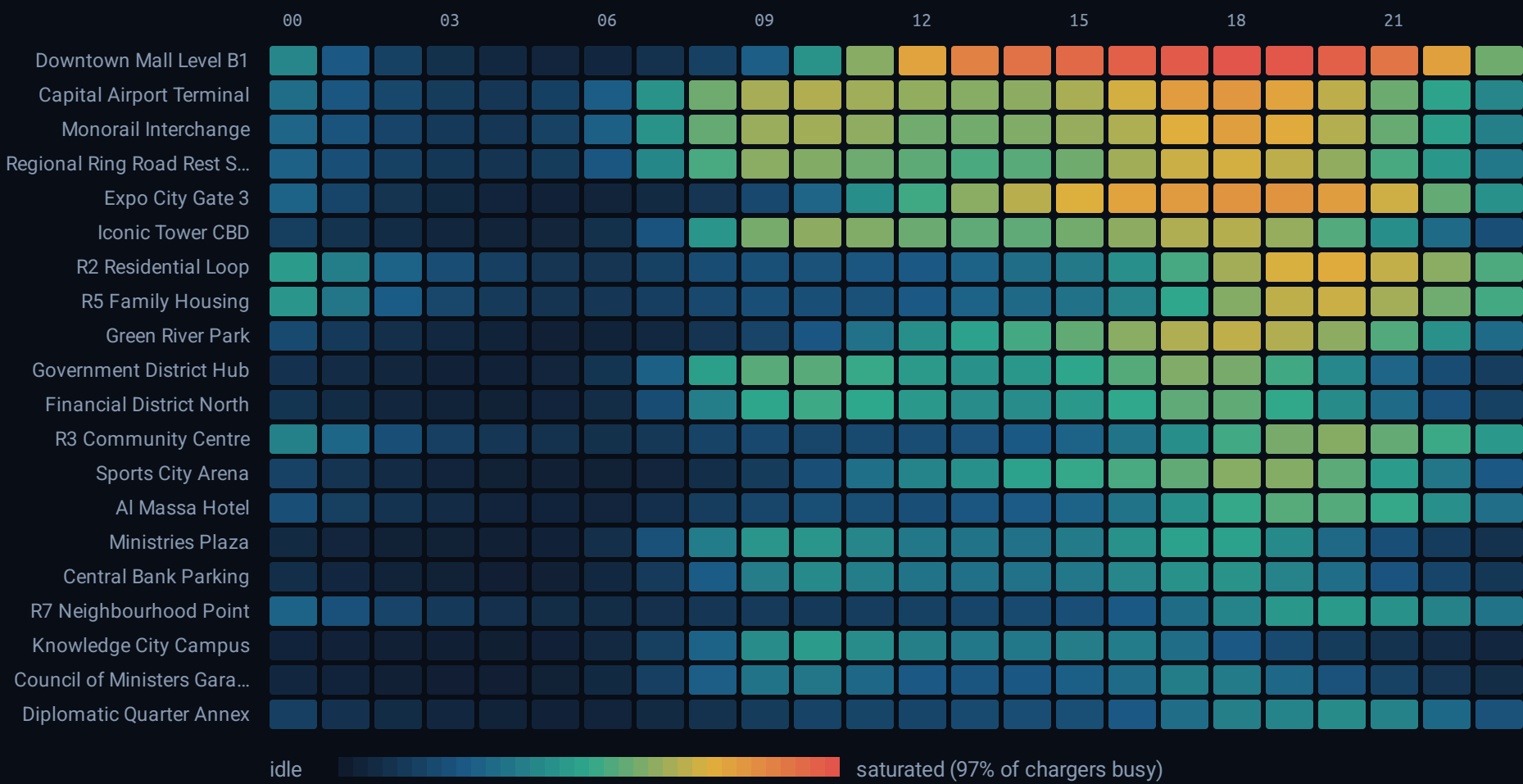
32.1 %

mean utilisation

3,439

kW city peak load

MEAN CHARGER UTILISATION — 20 STATIONS × 24 HOURS



The structure the models must learn: a commuter double-peak, a retail-led weekend, and stations that saturate while their neighbours sit idle.

Three models, one hour ahead

① arrivals

How many EVs plug in next hour?

0.746 -24 %
R² held out MAE vs baseline

② avg_wait_min

How long will a driver queue?

0.618 -41 %
R² held out MAE vs baseline

③ energy_kwh

How much grid energy is drawn?

0.927 -58 %
R² held out MAE vs baseline

VALIDATION – TIME SPLIT, LAST 2 MONTHS HELD OUT

TARGET	MAE	RMSE	R²	BASELINE	IMPROVEMENT
arrivals	0.980	1.505	0.746	same hour last week	-24.0 %
avg_wait_min	1.105	4.125	0.618	historical profile	-41.3 %
energy_kwh	14.52	23.48	0.927	same hour last week	-57.7 %

A random split would leak the future through the lag features. Training stops at 31 October; scoring runs on the 29,280 station-hours after it.

93.7 % 93.0 %
traffic-light accuracy predictions within 5 minutes

51 FEATURES, SIX FAMILIES

calendar & cyclical encodings · weather & scheduled events · station capacity · lags at 1/2/3/24/48/168 h · rolling means over 3/24/168 h · leak-free expanding (station × weekday × hour) profile · city-wide network state

WHAT THE QUEUE MODEL LEARNED

Its three strongest features are **arrivals**, **chargers per arrival** and **session length** — it rediscovered $\rho = \lambda / (c \cdot \mu)$ from queueing theory on its own.

Forecasting says what will happen. Optimisation decides what to do about it.

Once you know 30 cars arrive between 17:00 and 19:00, powering them is **not a prediction problem** — it is constrained allocation. So it is solved as a linear program, not another regressor.

minimise

$\sum \text{price}[t] \cdot x[v,t]$

energy bill

+ w · P_{peak}

flatten the profile

+ 500 · $\sum u[v]$

unmet energy

s.t.

$\sum x[v,t] + u[v] = E_v$

every car gets its charge

$0 \leq x[v,t] \leq P_v$

charger + vehicle limit

x = 0 outside [arrive, leave]

$\sum_v x[v,t] \leq \text{cap}[t]$

grid ceiling

Solved with HiGHS via `scipy.optimize.linprog`. 934 vehicles · 32,616 kWh requested. The unmet-energy slack keeps it always feasible.

AGGREGATE CITY LOAD OVER 24 HOURS



-28.3 %

peak demand
2,568 → 1,840 kW

92.7 %

energy delivered
on time — unchanged

4 → 0

hours breaching
the grid ceiling

Two products from one forecasting core

DRIVER APP near Downtown · 18:00 · battery 22 % → 85 %

Ranked by **total time** = drive + queue + charge — never by distance.

STATION	KM	DRIVE	QUEUE	CHARGE	TOTAL MIN
● Iconic Tower CBD	1.64	5.4	1.5	25.8	32.7
● Financial District North	1.03	3.9	1.4	43.0	48.3
● Government District Hub	1.34	4.7	2.2	43.0	49.9
● Downtown Mall Level B1	1.06	4.0	21.3	43.0	68.3
● Council of Ministers Garage	0.74	3.3	0.0	117.2	120.5

The system **rejects the two nearest stations**: the mall is 1 km away but carries a 21-minute predicted queue, and the ministries garage is empty but its 22 kW chargers would take two hours.

CITY OPERATOR DASHBOARD

- **24-hour outlook** — expected EVs, grid load, headroom against the ceiling
- **Congestion heatmap** — every station, every hour
- **Live network map** colour-coded by service level
- **Overload alerts** before the peak arrives

BUILT WITH

Python · pandas · scikit-learn · SciPy HiGHS · Streamlit · FastAPI

40

invariant tests passing

7

REST endpoints

4

dashboard views